

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY | GRADE 10 | SUB-STRAND 1.5: PERIODICITY | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Remember the driving question: Why do elements that sit next to each other in the periodic table, like sodium and magnesium, or chlorine and argon, behave so differently, and can we use an element's position in the table to predict exactly how it will behave before we ever put it in a test tube? Use what you observed in the Lake Victoria demonstration at the Kisumu Water and Sanitation Company and in Lessons 1 to 7. Write your answers in the space under each prompt. Use electron configurations in the form 2,8,1, and use the key terms nuclear charge, shielding, ionisation energy and full outer shell. Support each claim with evidence from the lessons. Write in full sentences, and include balanced equations where asked.

Part 1: The Kisumu Puzzle - Across Period 3

At the Kisumu Water and Sanitation Company, the water purification officer dropped sodium, magnesium and aluminium into Lake Victoria water. Sodium exploded. Magnesium barely fizzed. Aluminium showed no reaction with cold water.

Use the data below (from Lessons 5 and 6) and your own knowledge of electron configurations.

Element	Atomic number	Electron configuration	First ionisation energy (kJ/mol)
Sodium (Na)	11	2,8,1	496
Magnesium (Mg)	12	2,8,2	738
Aluminium (Al)	13	2,8,3	578
Argon (Ar)	18	2,8,8	1521

(a) Explain why sodium and magnesium, which are neighbours separated by one proton, react so differently with cold water. Use the ideas of nuclear charge, shielding and the first ionisation energy values in the table. (b) Describe the trend in reactivity with cold water across Period 3 from sodium to

aluminium, and explain it. Aluminium's first ionisation energy is lower than magnesium's; use your knowledge of electron configurations to suggest why this dip happens, and say what other factor you think explains why aluminium still shows no reaction with cold water. (c) Write a balanced equation with state symbols for the reaction of sodium with water, and state the observation from the Lake Victoria demonstration that shows the solution is alkaline.

Write your answer here:

Part 2: Going Down Group I

In Lessons 1 and 2 you saw lithium, sodium and potassium react with cold water. Lithium reacted slowly and steadily, sodium reacted faster, and potassium reacted most violently with a lilac flame.

(a) Write the electron configurations of Li, Na and K. (b) Explain why potassium is more reactive than sodium even though potassium has many more protons. Your explanation must refer to the number of electron shells, atomic radius, shielding and ionisation energy. (c) Name the two products of the reaction of potassium with water and describe how you would prove that one of them is a gas.

Write your answer here:

Part 3: Two Families, Opposite Trends

In Lesson 4 you studied Group VII, the halogens: chlorine (Cl_2 , yellow-green gas), bromine (Br_2 , orange-brown liquid) and iodine (I_2 , grey-black solid). Chlorine water displaced bromide ions from potassium bromide solution, and bromine water displaced iodide ions from potassium iodide solution.

(a) Write a word or balanced equation for the displacement reaction of chlorine water with potassium bromide solution. (b) Compare how reactivity changes down Group I with how it changes down Group VII. (c) Explain why the two trends go in opposite directions, even though both are caused by more electron shells. (d) Explain why the Kisumu water officer uses chlorine rather than bromine or iodine to treat Lake Victoria water.

Write your answer here:

Part 4: The Inert Shield

The officer transported the sodium in a sealed ampoule of argon gas. Argon sits right next to the very reactive chlorine, yet it reacts with nothing. Use the table in Part 1 and your knowledge of electron configurations.

(a) Compare the first ionisation energy of argon with that of sodium. How many times larger is the argon value, approximately? (b) Explain why argon is completely inert, referring to its outer shell. (c) Explain why argon, rather than ordinary air, was needed to protect the sodium.

Write your answer here:

Part 5: Final Explanation - Predicting from Position

Now answer the driving question in a single connected explanation: Why do neighbouring elements such as sodium and magnesium, or chlorine and argon, behave so differently, and can we use an element's position in the periodic table to predict how it will behave before we put it in a test tube?

Your answer must: (1) explain what an element's group and period tell you about its electron configuration; (2) explain the Group I down-group trend and the Period 3 across-period trend using shielding, nuclear charge and ionisation energy; (3) use at least one example of a Kenyan application of an element and link it to a periodic property; (4) give a prediction for how a new element would behave based on its position, and justify it. Choose either calcium (Group II, 2,8,8,2) compared with magnesium, or rubidium (below potassium in Group I).

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of electron configuration and position	Correctly writes electron configurations and clearly links group and period to outer electrons and shells in every part.	Writes mostly correct configurations and links position to electrons in most parts, with minor gaps.	Configurations are missing or incorrect, and position is not clearly linked to electrons.
Explanation of trends using nuclear charge, shielding and ionisation energy	Accurately explains the Group I, Period 3 and Group VII trends, using all three ideas and showing why Group I and Group VII trends are opposite.	Explains most trends with two of the key ideas, with some imprecision.	Describes trends without a causal explanation, or uses ideas incorrectly.
Use of evidence and data	Cites specific observations and values (such as 496, 738 and 1521 kJ/mol, lilac flame, kanga bleaching) accurately and consistently.	Cites some evidence accurately, though not always specifically.	Little or no evidence cited, or the evidence is inaccurate.
Equations and chemical accuracy	Balanced equations with correct state symbols, correct products and correct tests.	Equations mostly correct, with minor errors in balancing or state symbols.	Equations incorrect or missing.
Application, prediction and connection to the phenomenon	Gives a justified prediction and a Kenyan application, and returns to the Kisumu phenomenon to answer the driving question completely.	Gives a reasonable prediction and application with a partial link to the phenomenon.	Prediction or application is missing or unjustified, with no clear link back to the phenomenon.

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